

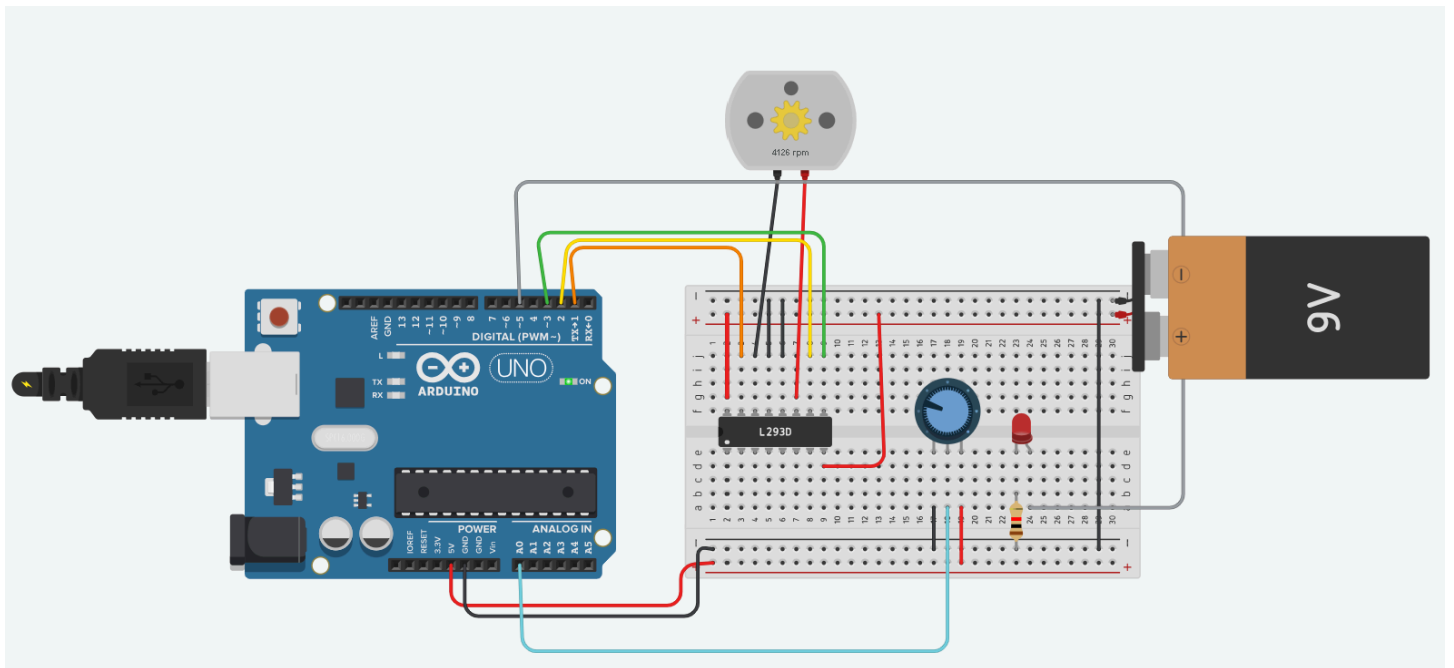
SRM TEAM

ROBOCON

Created by Rayan Raphy

Assignment -1

Circuit Diagram



Source Code

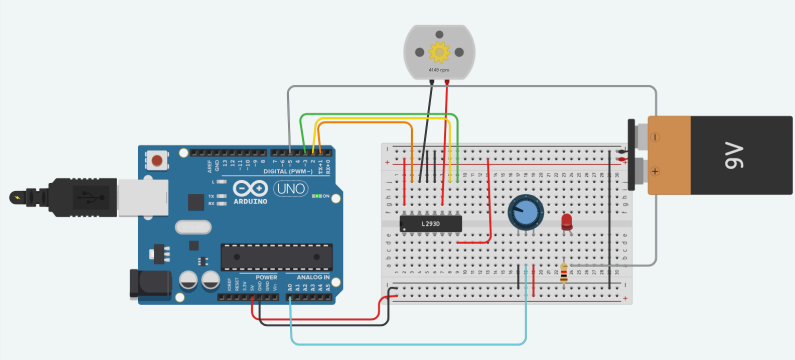
```
1  /*
2
3
4  code By Rayan Raphy
5  */
6
7  //INitializine the pins!!
8  const int in1 = 1;
9  const int in2 = 2;
10 const int en = 3;
11
12 // just a additional speed indicator
13 const int led =5;
14
15 const int pot = A0;
16
17 void setup()
18 {
19     //setting up the out nad input pins
20     pinMode(in1, OUTPUT);
21     pinMode(in2, OUTPUT);
22     pinMode(en, OUTPUT);
23
24     pinMode(led, OUTPUT);
25
26     pinMode(pot, INPUT);
27     Serial.begin(9600);
28 }
29
30 void loop()
31 {
32     int potVal = analogRead(pot); //reading the potentiometer values
33     int speed = map(potVal,0,1023,0,255); // mapping the potentiometer values
34
35     //Setting the direction of rotation
36     digitalWrite(in1,LOW);
37     digitalWrite(in2,HIGH);
38
39     //changing the speed according to the potentiometer
40     analogWrite(en,speed);
41     analogWrite(led,speed);
42
43 }
```

OUTPUT SCREENSHOT!!

Simulator time: 00:00:07

Code Stop Simulation Send To

1 (Arduino Uno R3)



```
24 const int en = 3;
25 //additionally pseed viewing features
26 const int led = 5;
27
28 const int pot = A0;
29
30 void setup()
31 {
32   pinMode(in1, OUTPUT);
33   pinMode(in2, OUTPUT);
34   pinMode(en, OUTPUT);
35
36   pinMode(led, OUTPUT);
37
38   pinMode(pot, INPUT);
39
40 }
41
42 void loop()
43 {
44   //acquiring value from potentiometer
45   int potVal = analogRead(pot);
46   int speed = map(potVal,0,1023,0,255);
47
48   //setting up the direction
49   digitalWrite(in1,LOW);
50   digitalWrite(in2,HIGH);
51   //setting up the speed
52   analogWrite(en,speed);
53   analogWrite(led,speed);
54
55 }
```

Serial Monitor